

AICoach: A System Framework for Online Realtime Workout Coach

Haocong Ying, Tie Liu*, Mingxin Ai, Jiali Ding, Yuanyuan Shang

Capital Normal University

Beijing, China

yhc.pro@outlook.com, liutiel@163.com, a970052666@163.com, sy_djl@163.com, yshang@cnu.edu.cn

ABSTRACT

There is a growing demand for online fitness due to the impact of the epidemic. This paper presents a real-time online fitness system framework called AICoach, which offers different online coaches. The framework constructs an extensible AI-based architecture that supports a variety of fitness movements. Firstly, key frames of motion are extracted automatically, and the feature vectors are calculated with the body pose points. Secondly, the state transition matrix can effectively identify fitness actions and capture their time-continuous characteristics. Finally, AICoach can accurately provide the number of repetitions and correction tips of fitness movements. Currently, the AICoach has a number of fitness courses supported by online coaches and has been tested on hundreds of fitness movements. The code can be downloaded from <https://github.com/liu>

CCS CONCEPTS

• Software and its engineering → Software libraries and repositories; • Information systems → Multimedia information systems; • Computing methodologies → Computer vision.

KEYWORDS

Fitness Action Recognition, Fitness Repetition Counting, Fitness Action Correction

ACM Reference Format:

Haocong Ying, Tie Liu*, Mingxin Ai, Jiali Ding, Yuanyuan Shang. 2021. AICoach: A System Framework for Online Realtime Workout Coach. In *Proceedings of the 29th ACM International Conference on Multimedia (MM '21)*, October 20–24, 2021, Virtual Event, China. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3474085.3478321>

1 INTRODUCTION

With the wide spread of COVID-19, fitness centres have been closed around the world. However, people's demand for fitness has not decreased but increased. It is sensible to fitness at home with a virtual fitness coach. A virtual fitness coach system should meet the following requirements: a) The system should have various sets of carefully designed curriculums for users to accomplish; b) The system can track the users' process of accomplishing the curriculums;

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '21, October 20–24, 2021, Virtual Event, China

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8651-7/21/10... \$15.00

<https://doi.org/10.1145/3474085.3478321>

c) The system can correct the users' actions. Besides, the count number should be as precise as possible while the cost should be relatively low. Last but not least, the system should be able to run in real time.



Figure 1: The application scenarios of AICoach: workout repetition counts and correction tips.

Virtual coaches should be aware that counting the number of repetitions of fitness actions is the core function of the system. With the advance of hardware technology, wearable sensor based methods are becoming popular in recent years [7]. The sensors can perform human pose tracking with high robustness while they are too expensive to deploy. However, in the fitness scene, the sensors would affect the fitness actions and could probably cause accidents. Methods utilizing 3D cameras [10] or multiple cameras [8] can also generate robust counting results, but they have the same problems in cost and deployment as the sensor-based methods.

When it comes to the computer vision based methods with single camera, people traditionally estimate a personal movement by extracting optical flow trajectories from video frames [4]. However, the method based on optical flow may fail in large-scale and fast motion, and the calculation of optical flow requires a large amount of computing resources.

OpenPose [2] has been introduced in 2018, a method which extracts skeleton points of a person from an image based on the convolutional neural network. Simply send an image or a video frame into the neural network, it would return a heatmap to predict the locations of the skeleton points. However, OpenPose requires high performance devices such as desktop-level GPUs. To inference in real time on relatively affordable devices, a redesigned OpenPose implementing MobileNet principles is proposed[3][5], and the inference speed has been raised to the real-time level.

Based on the skeleton points extracted from the pose detector, a method[6] has been proposed for counting action repetitions by evaluating the similarities between each two frames in the video by

calculating the sum of Euclidean distances of corresponding skeleton points. This is a method only for integrated recorded video clips instead of real-time video stream captured from webcams, and the human body structure is ignored when measuring similarities between the frames with Euclidean distances.

A method [1] has been proposed to count action repetitions by calculating the sum of the cosine value of selected human joint angles with the aid of OpenPose. When the sum of the value reaches a high threshold and a low threshold in sequence by handcraft, the system counts once. However, the joint angles are not robust features since the aspect ratio of the input video could affect it a lot, and this method could not support counting complex actions with more parts of body involved and more than two key frames.

Here we propose AICoacher, a real-time virtual fitness coach system based on the skeleton points extracted from OpenPose. The system has a set of curriculums for the users to exercise, and all fitness actions inside the schedule have count limits or time limits. The system could count human action cycles in real scene in real time with only one camera. A timer is also included in the system, so that the system could shift to the next fitness action when the current action reaches the count limit or the time limit. Besides, the proposed system can also correct the fitness actions by notifying the user the correct fitness action.

2 SYSTEM FRAMEWORK

2.1 Architecture

The overall architecture of AICoacher is illustrated in Figure 2. In AICoacher, the frames for processing are obtained by the VideoIO module. The PoseDetector module is designed to extract skeleton points and their confidence scores from each of the video frames. The AIClass module manages the schedule, the limits of counts and time of all actions, and the HealthAction module is designed together with the ActionCounter and the ActionTimer to manage the count and time data of the current action. The ActionTransfer module manages an internal state machine to track the transfers between the key frames with the assistance of the WorkoutParameter module which defines the method of generating features for each of the actions. The ActionTransfer module also checks the correctness of the performed actions with the WorkoutParameter module which has the features of the correctly performed actions defined in it.

In designing AICoacher, we focus on the system's extensibility. The definitions and arrangements of the actions can all be edited in .txt files instead of being hard coded, making this system highly customizable for counting and correcting different types of actions.

2.2 Process

The pipeline of our system is illustrated in Figure 3, which shows the processing pipeline of each input frame.

Firstly, a frame from a real-time webcam or a video file is captured by the VideoIO module. Then the PoseDetector module based on OpenPose is applied to extract the skeleton points and their confidence scores from the frames. This process is followed by calculating a feature vector of the frame based on the positions of the skeleton points according to the algorithms defined in the WorkoutParameter module. Then, the ActionTransfer would compare

the generated feature vectors with the features of the next key frame to decide whether the current frame reaches the next key frame. The ActionTransfer module would also check whether the current key frame is accomplished correctly by comparing the calculated feature vectors with the feature vectors of the correctly performed the current key frame.

The ActionCounter would count once when a cycle of key frames is completed. When the system counts, the ActionCounter would refresh the count value, and when every frame is processed, the ActionTimer would refresh the time data. The whole processing mechanism would stop when the above modules find either of them reaching the limit.

2.3 AIClass Module: Manage the Schedule of Actions

While designing AICoacher, we call the frames containing a sequence of cycled features an 'action'. To support the calculation of more than one kind of actions according to the user's arrangement and set the limit of the number of actions or action time, we design the AIClass module to manage the actions.

While being initialized, the AIClass module loads the information of all actions from a text file in which the unique ID, the name, the count limit and the time limit of every action are defined. When an action is selected, the HealthAction module would be initialized according to the unique ID with the above information provided by AIClass module.

2.4 PoseDetector Module

When a frame is captured by the VideoIO module from a real time webcam or a video file, the PoseDetector module based on OpenPose would extract the skeleton points from the frame. In the input end, a image resized to 192*192 should be fed into the neural network, and the neural network would output a heatmap matrix with the size of (92*92)*14, indicating the probability distribution of the 14 skeleton points on the pixels. Then the module tries to find the maximum value in each column of the matrix to ensure the most possible place of the skeleton points and the confidence value of the location. Finally, the extracted locations of skeleton points would be scaled to the aspect ratio of the original input frame, so that the system would get rid of the interfere from the input aspect ratios.

2.5 ActionTransfer Module and PoseParameter Module

The ActionTransfer module manages an internal state machine which labels the position of the current frame in the whole key frame cycle of the action. The ActionTransfer module labels the current key frame whatever it is already correctly performed. Last but not least, the ActionTransfer module is initialized as soon as an HealthAction module is initialized. They are both initialized according to the action information provided by the AIClass module.

The ActionTransfer module would load the parameters of all key frames of the current action from a text file. In the text file, the first lines contain the name of the action, the unique ID of the action and the number k of key frames of the action. In the following k lines of the text file, the feature vectors of the key frames are all

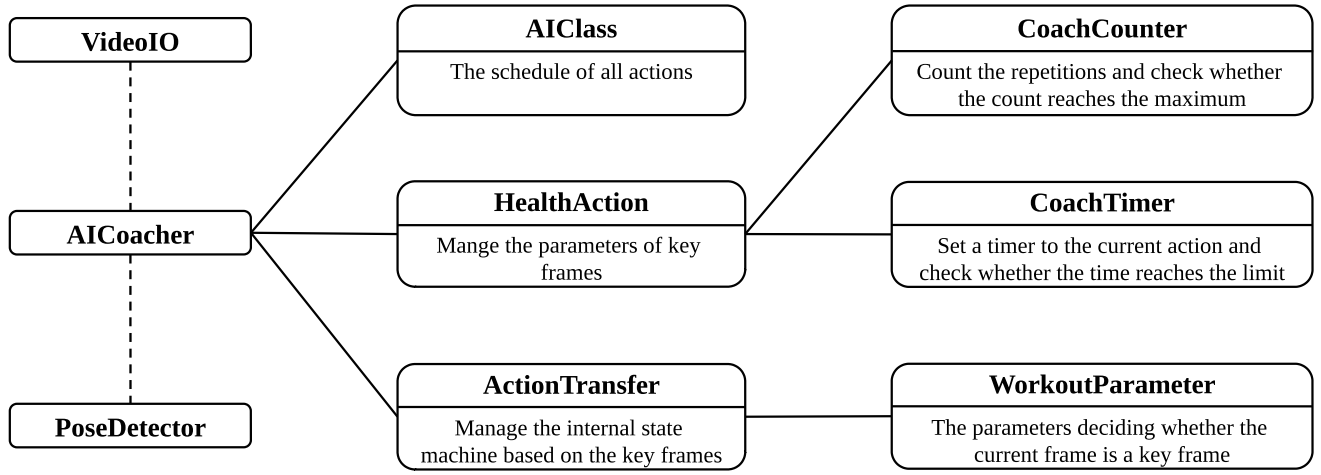


Figure 2: The overall architecture of AICoacher.

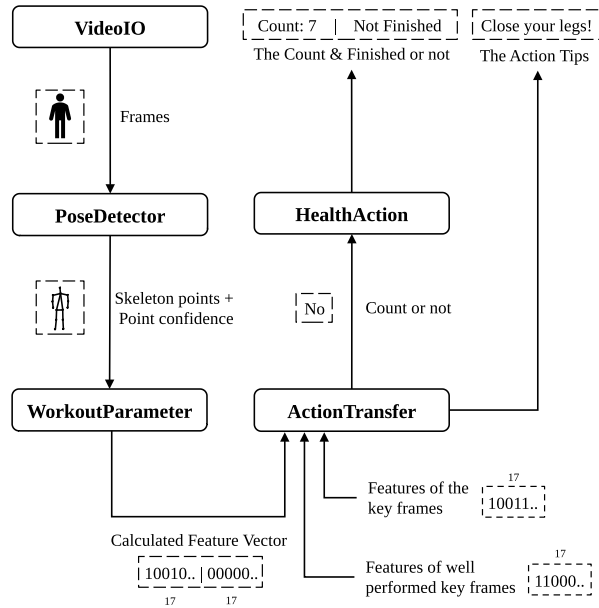


Figure 3: The frame processing steps of AICoacher.

defined. Each line here refers to a key frame, while the first seventeen columns in the line refer to the features for recognizing the key frame and the second seventeen columns refer to the features of the strictly performed key frame.

When a set of skeleton points have been extracted by the PoseDetector module, the features are calculated by the WorkoutParameter module, forming a vector whose digit is a calculation result of a feature. Then each digit of the feature vector would be compared with the next key frame. If digit in the first seventeen columns of those two vectors are identical, the key frame would be seen as

completed and the label in ActionTransfer module would jump forward to the next frame. At the same time, the correctness of the key frame would be checked by comparing the last seventeen columns of the feature vector, where lies more strict trigger standards. If the next key frame is reached without executing the previous key frame, a notification would be sent to the user, showing a guidance to correctly perform the action.

If the whole cycle of key actions has been accomplished, the key frames' label go back to the first key frame of the action and the ActionCounter module affiliated to the HealthAction module would count once.

3 EVALUATION

The core feature of the entire virtual fitness coach is human action counting. While designing the system, two metrics should be taken into consideration: correctness and performance. Thus, we have tested the system according to the two metrics.

To measure the correctness of the action counting module of the proposed system, we have tested it on 20 videos with real people under real scene, and they perform different types of fitness actions that AICoacher supports. Among the videos for testing, people performed 105 repetitions of 6 categories of actions and AICoacher counts 102 times, meaning that the system could work robustly even under relatively complicated scenes.

To further ease the computing burden, we have designed a soft real-time mechanism so that the system obtains frames only need 0.1s once. Thus, the system would run smoothly even on devices without any xPU acceleration with almost no correction rate drops.

4 DEPLOYMENT AND USAGE

4.1 Availability and Deployment

AICoacher is released under the license of Apache 2.0 and its code is openly available at: <https://github.com/liutiel/AICoacher>. We also

provide extensive documentations and sampled projects for it. Contributions from the open-source community are welcome, via the GitHub issues/pull request mechanisms.

In the repository, we provide a binary release of AICoacher which can run directly on x64 platforms of Windows. The source code is recommended to be compiled and run on Windows 10 x64 platform with Visual C++ 2019 installed. The GUI of AICoacher is designed under Qt 5.15.2 which is also required when compiling AICoacher, and OpenCV 4.5.1 is utilized in the system in order to process the input and output video streams. Besides, a third-party TensorFlow-Lite C++ interface should be installed following the introduction at [https://github.com/ValYouW/tflite-dist\[9\]](https://github.com/ValYouW/tflite-dist[9]) to run the OpenPose with C++.

4.2 Interfaces and Usage

Based on the above modules, three interfaces have been designed for AICoacher:

Table 1: Interfaces

Interface	Description
AICoacherInitialize	Initialize AICoacher with specified class schedule and specified input video stream.
AICoacherStart	Start fitness training.
AICoacherStop	Stop fitness training.
AICoacherSuspend	Suspend fitness training.
AICoacherResume	Resume fitness training from being suspended.

Five types of messages would be sent by the AICoacher library to the GUI:

Table 2: Handles

Output Messages	Description
MESSAGE_COUNT	Count of the repetitions of the current action.
MESSAGE_TIME	Duration of the current action.
MESSAGE_PROMPT	If the user have not done an action correctly, a prompt would be sent to the GUI.
MESSAGE_CLASS	Name of the current fitness class.
MESSAGE_ACTION	Name of the current fitness action.

Based on the interfaces and messages, we have developed a GUI for end-user as shown in Figure 4, where both real-time video stream and video files could be set as input stream. The curriculum name, the action name, the count of the current action, the duration time of the current action and the action tips for the user are all displayed on the GUI. Users can choose curriculums freely from the dropdown menu, and can start, stop, pause or resume training at any time.

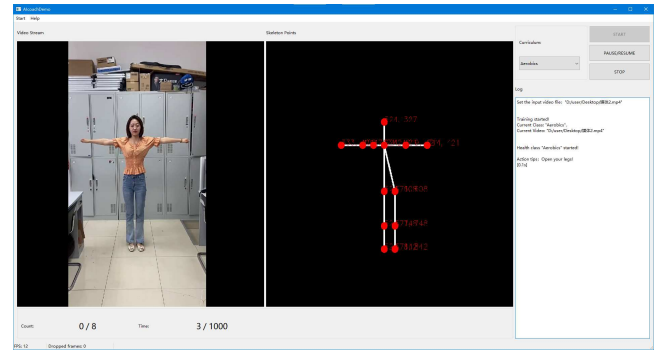


Figure 4: The system GUI of AICoacher.

5 CONCLUSION

AICoacher presents and implements an online real-time workout coach system framework. The system framework constructs an extensible AI-based architecture that supports a variety of fitness movements. The key frames of one fitness action are extracted automatically and the feature vectors are calculated with the body pose points. The fitness actions are efficiently recognized with a state transition matrix while the temporal continuous characteristics of one fitness action can be well captured. The fitness repetition counts and correction tips can be well output by AICoacher so that the user can be instructed for the better fitness experience. AICoacher can be extended to other action recognition tasks just like dance teaching, behavior correction and other human-computer interaction applications.

REFERENCES

- [1] Talal Alataiah and Chen Chen. 2020. Recognizing Exercises and Counting Repetitions in Real Time. *arXiv preprint arXiv:2005.03194* (2020).
- [2] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. 2019. OpenPose: realtime multi-person 2D pose estimation using Part Affinity Fields. *IEEE transactions on pattern analysis and machine intelligence* 43, 1 (2019), 172–186.
- [3] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, et al. 2019. Searching for mobilenetv3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1314–1324.
- [4] Rushil Khurana, Karan Ahuja, Zac Yu, Jennifer Mankoff, Chris Harrison, and Mayank Goel. 2018. GymCam: Detecting, recognizing and tracking simultaneous exercises in unconstrained scenes. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 4 (2018), 1–17.
- [5] Ildoo Kim. 2019. Deep Pose Estimation implemented using Tensorflow with Custom Architectures for fast inference. *GitHub*. URL: <https://github.com/ildoonet/tf-pose-estimation> [accessed 2020-12-12] (2019).
- [6] Nikolay Neshov, Agata Manolova, Krasimir Tonchev, and Ognian Boumbarov. 2019. Detection and Analysis of Periodic Actions for Context-Aware Human Centric Cyber Physical System to Enable Adaptive Occupational Therapy. In *2019 10th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS)*, Vol. 2. IEEE, 685–690.
- [7] Andrea Soro, Gino Brunner, Simon Tanner, and Roger Wattenhofer. 2019. Recognition and repetition counting for complex physical exercises with deep learning. *Sensors* 19, 3 (2019), 714.
- [8] Kosuke Takahashi, Dan Mikami, Mariko Isogawa, and Hideaki Kimata. 2018. Human pose as calibration pattern; 3D human pose estimation with multiple unsynchronized and uncalibrated cameras. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 1775–1782.
- [9] ValYouW. 2021. tflite-dist. <https://github.com/ValYouW/tflite-dist>.
- [10] Xiaolin Wei, Peizhao Zhang, and Jinxiang Chai. 2012. Accurate realtime full-body motion capture using a single depth camera. *ACM Transactions on Graphics (TOG)* 31, 6 (2012), 1–12.